

Association of elevated neutrophil-to-lymphocyte ratio with increased intracranial aneurysm stability scores and aneurysm growth

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Background and purpose: Inflammation involves in the progression of intracranial aneurysms (IAs). However, whether the neutrophil-to-lymphocyte ratio (NLR) as an inflammatory marker links to IAs stability is unidentified. This study was performed to assess the association of the NLR with IAs stability. **Methods:** We retrospectively reviewed the medical records of patients diagnosed with unruptured IAs from January 2014 to June 2018. According to the quartiles of the NLR, patients with unruptured IAs were categorized into four groups. We evaluated the association between the NLR and IAs stability scores and IAs growth. Multiple logistic regression models were used in the analysis. **Results:** A significant dose-response association was found between the NLR with IAs stability scores and IAs growth. After adjustment for potential confounders, an elevated NLR (fourth quartile) was associated with increased PHASES score (>5) (adjusted odds ratio [OR], 2.007; 95% confidence interval [CI], 1.361–2.960; $p < 0.001$ [p for trend <0.001]), increased ELAPSS score (>15) (adjusted OR, 1.581; 95% CI, 1.074–2.328; $p = 0.020$ [p for trend =0.001]), increased JAPAN 3-year rupture risk score (>5) (adjusted OR, 1.512; 95% CI, 1.033–2.215; $p = 0.034$ [p for trend <0.001]), and IAs growth (adjusted OR, 16.759; 95% CI, 3.022–92.928; $p = 0.001$ [p for trend <0.001]). **Conclusion:** An elevated NLR was associated with increased IAs stability scores and IAs growth. The association between NLR and IAs stability need further investigate.

Keywords: Neutrophil-to-lymphocyte ratio—Intracranial aneurysm—Inflammation—Stability

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Introduction

Intracranial aneurysms (IAs) affect approximately 2% to 4% of the adult population.^{1–3} If the IAs ruptures, the mortality is as high as 30–40%.^{4,5} Therefore, screening unstable IAs for appropriate intervention is crucial to reduce the death risk of IAs. Several growth and rupture risk scores for predicting the stability of unruptured IAs have been established, including the PHASES score,⁶ ELAPSS score,² and JAPAN 3-year rupture risk score.⁷ Although the biological factors contributing to aneurysm growth remain unclear, aneurysm growth is pathologically associated with destruction of the inner elastic layer secondary to impairment of vascular smooth muscle cells, which further leads to disordered aneurysm wall collagen synthesis and extracellular matrix remodeling.^{8,9}

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Long-term clinical observations have revealed that several leukocytes, including neutrophils, lymphocytes, and monocytes/macrophages, infiltrate the adventitia of aneurysms and have potential roles in the pathophysiology of IAs.¹⁰ The presence of neutrophil-derived enzymes in the aneurysm wall is associated with a significant increase in the risk of rupture.¹¹ Recent studies have shown that neutrophil extracellular traps promote the development of IAs rupture.¹² Lymphocytes have been shown to play an important role in inflammatory diseases associated with aneurysms, such as atherosclerosis.^{13,14} Lymphocytes can promote extracellular matrix remodeling and phenotypic regulation of vascular smooth muscle cells, thereby affecting the development of IAs.¹⁵ In addition, the latest research shows that subsets of circulating leukocytes, such as neutrophils and lymphocytes, are associated with the development of abdominal aortic aneurysms.¹⁶

The neutrophil-to-lymphocyte ratio (NLR) is a measure of the proportion of systemic neutrophils and lymphocytes, and this ratio has been considered a reliable composite marker and dynamic index of systemic inflammation.^{17,18} However, few data on the relationship between the NLR and aneurysm stability are available. In addition, NLR is convenient in practical application because neutrophil and lymphocyte counts are routinely measured during laboratory testing. NLR can provide relatively cheap diagnosis for clinical treatment.

This study was performed to investigate the relationship of the NLR with the stability and growth of unruptured IAs.

Methods

Study design and patients

We identified unruptured IAs patients, older than 18 years, were consecutively admitted to Beijing Tiantan Hospital, a large comprehensive stroke center located in Beijing, China, from January 2014 to June 2018. The diagnosis of unruptured IAs was confirmed by CTA, MRA, and/or DSA. We excluded cerebral AVM, AVF, or moyamoya disease related aneurysms, as well as dissecting, fusiform, traumatic, and mycotic aneurysms, and patients with systematic inflammatory, infectious diseases, or cancer. Ethics approval was obtained from the hospital review committee. All participants provided informed consent. This study complies with the principles of the Declaration of Helsinki.

Data collection

We reviewed the electronic medical records and integrated the demographic, clinical, and laboratory data into our neurovascular database. We collected the patients' demographics and medical history, including subarachnoid hemorrhage, ischemic stroke, hypertension,

hyperlipidemia, diabetes mellitus, heart disease, smoking status, and drinking status. The laboratory data were collected after patient admission. Blood samples were tested as follows: First, venous blood samples were collected by venipuncture and placed in K2-EDTA anticoagulation tubes. Second, blood samples were analyzed within 6 h by an autoanalyzer (Mindray Hematology Analyzer BC-6900 series; Mindray Corporation, Shenzhen, China) in the hospital. The reference ranges of the assay were as follows: white blood cells, 4.0-10.0 $10^9/L$; neutrophils, 2.0-7.5 $10^9/L$; and lymphocytes, 0.9-5.2 $10^9/L$. Multiple blood tests were performed during the imaging follow-up, and the mean of the blood sample analysis results were used for data assessment.

Image analysis and aneurysm stability evaluation

All images were read by two readers who had 4 and 5 years of experience in vascular neuroimaging, respectively, and who were blinded to each other's interpretations and the patients' clinical information. Before the image evaluation, we conducted Kappa test ($\kappa > 0.8$), which indicated that the results of aneurysm characteristics judged by the two readers were high consistency. In the process of image evaluation, discordances over the evaluation results between the two readers were resolved through discussion, with the involvement of senior radiologist with 10 years of experience when necessary. Discordances were defined as: (1) For categorical variables, inconsistent discrimination of the shape, and aneurysmal location; and (2) for the continuous variables, such as aneurysm size, neck width, maximum height, maximum width, parent vessel diameter, the measurement difference between the two readers is greater than 1 mm.

Aneurysm stability was evaluated by the PHASES score,^{1,6} ELAPSS score,² and JAPAN 3-year rupture risk score.⁷ The PHASES score for each aneurysm was calculated by 6 factors: population (0, North American, European (other than Finnish); 3, Japanese; 5, Finnish), hypertension (0, no; 1, yes), age (0, <70 years; 1, ≥ 70 years), earlier SAH from another aneurysm (0, no; 1, yes), size of aneurysm (0, <7.0 mm; 3, 7.0–9.9 mm; 6, 10.0–19.9 mm; 10, ≥ 20.0 mm), aneurysm site (0, internal carotid artery; 2, middle cerebral artery; 4, anterior cerebral arteries, posterior communicating artery, and posterior circulation).¹⁹ ELAPSS score is based on 6 factors: earlier subarachnoid hemorrhage (no; yes), location of the aneurysm (ICA / ACA / ACOM; MCA; PCOM / Posterior circulation), age (>60 years (per 5 years)), population, size of the aneurysm (1.0-2.9; 3.0-4.9; 5.0-6.9; 7.0-9.9; ≥ 10 (mm)), and shape of the aneurysm (regular; irregular).²⁰ JAPAN 3-year rupture risk score has six predictors as follows: age, sex, hypertension, aneurysmal size, aneurysmal location, and presence of daughter sac.⁷ According to the risk of aneurysm rupture or growth within 5 or 3 years, we classified unstable aneurysms as those with a PHASES

score of >5, ELAPSS score of >15, and JAPAN 3-year rupture risk score of >4. We used PHASES, ELAPSS, and JAPAN 3-year rupture risk score to predict aneurysm growth because their scoring items and structure are similar, and previous studies have also shown that the PHASES scores can predict the growth of aneurysms.¹

Follow-up imaging and aneurysm growth

Follow-up imaging was performed at 6 and 12 months and then annually for 48 months after diagnosis. Aneurysm growth was defined according to the following three criteria: growth of ≥ 1.0 mm in at least one direction as shown by identical or different imaging modalities, growth of ≥ 0.5 mm in two directions as shown by identical imaging modalities; or an indisputable change in aneurysm shape (i.e., change from regular shape to irregular shape).¹ The imaging follow-up time for all aneurysms was at least 6 months.

Statistical analysis

The baseline variables were compared between the groups using the χ^2 test or Fisher's exact test for categorical variables and one-way analysis of variance or the Kruskal–Wallis test for continuous variables with a normal or skewed distribution, respectively. Continuous variables are expressed as mean $\pm$ standard deviation or median with interquartile range. Categorical data are presented as proportions.

We took the fourth quintile with the lowest incidence of outcomes as a reference and performed logistic regression to explore the associations of the NLR with the aneurysm stability score and aneurysm growth. Adjusted odds ratios (ORs) and their 95% confidence intervals (CIs) were calculated. We used two models. In the first model, no variable adjustment was performed. In the second model, we included all potential confounding variables.

In addition, we used restricted cubic splines to examine the shape of the association of the NLR with the aneurysm stability score and aneurysm growth, with four knots (at the 25th, 50th, 75th, and 95th percentiles); the reference point for the NLR was NLR=2, and the OR was adjusted for all potential covariates (Model 2).

Additionally, we also used the C statistic, net reclassification index (NRI), and integrated discrimination index (IDI) to evaluate the incremental predictive value of the NLR beyond the PHASES score, ELAPSS score, JAPAN 3-year rupture risk score, conventional model, and aneurysm growth. With reference to previous studies, the risk category for aneurysm growth was set at <5%, 5% to <10%, and >10%.^{1,2} Categorical net reclassification was evaluated by cross-tabulating the participants into three categories of <5%, 5% to <10%, and >10% before and after addition of the NLR to the predictive model.

All statistical analyses were performed using R software version 3.6.3 (R Foundation for Statistical

Computing, Vienna, Austria). A two-tailed p value of <0.05 was considered statistically significant.

Results

Characteristics of the cohort

In total, 1349 consecutive patients with unruptured IAs were enrolled in this study, and 937 patients with 1088 unruptured IAs were ultimately analyzed after application of the exclusion criteria (Fig. 1). The baseline characteristics of the included and excluded patients were presented in Supplemental Table 1.

Among these 1088 unruptured IAs, the mean age at diagnosis was 56.0 ± 10.0 years, and 714 (65.6%) patients were women. As shown in Supplemental Table 2, advanced age ($p < 0.001$), subarachnoid hemorrhage ($p < 0.001$), hyperlipidemia ($p < 0.001$), diabetes mellitus ($p = 0.010$), and heart disease ($p = 0.003$) were more common in the aneurysm imagine follow-up group. The aneurysm size ($p < 0.001$), maximum height ($p < 0.001$), neck width ($p < 0.001$), maximum width ($p < 0.001$), aspect ratio ($p = 0.002$), size ratio ($p < 0.001$), and parent vessel diameter ($p = 0.034$) were smaller in the aneurysm imagine follow-up group than aneurysm stability score group. The aneurysms in the aneurysm stability score group more often had a daughter sac ($p = 0.003$). In addition, the aneurysms in aneurysm imagine follow-up group were mostly located in the internal carotid artery or middle cerebral artery ($p < 0.001$); and had a lower PHASES score ($p < 0.001$), ELAPSS score ($p < 0.001$), and JAPAN 3-year rupture risk score ($p < 0.001$). The median NLR was 1.82 (interquartile range, 1.43–2.40).

Relationship between aneurysm stability score and NLR

The median NLR in the aneurysm stability score group was 1.79 (interquartile range, 1.42–2.39). The baseline characteristics of the aneurysms in aneurysm stability score group by NLR quartiles are shown in Table 1.

In the aneurysm stability score group, 356 (39.9%) had a PHASES score of >5 points, 388 (43.5%) had an ELAPSS score of >15 points, and 389 (43.6%) had a JAPAN 3-year rupture risk score of >4 points.

Table 2 shows the association between NLR quartiles and aneurysm stability scores. Aneurysms in the higher NLR categories were associated with a PHASES score of >5 points (p for trend <0.001), ELAPSS score of >15 points (p for trend =0.001), and JAPAN 3-year rupture risk score of >4 points (p for trend <0.001). Compared with the lowest quartile, the second, third, and fourth NLR quartiles were associated with an increased risk of the following: PHASES score of >5 points (adjusted OR, 1.267 [95% CI, 0.865–1.855], 1.490 [95% CI, 1.018–2.183], and 2.007 [95% CI, 1.361–2.960], respectively), ELAPSS score of >15 points (adjusted OR, 1.284 [95% CI, 0.874–1.887], 1.045 [95% CI, 0.709–1.539], and 1.581 [95%

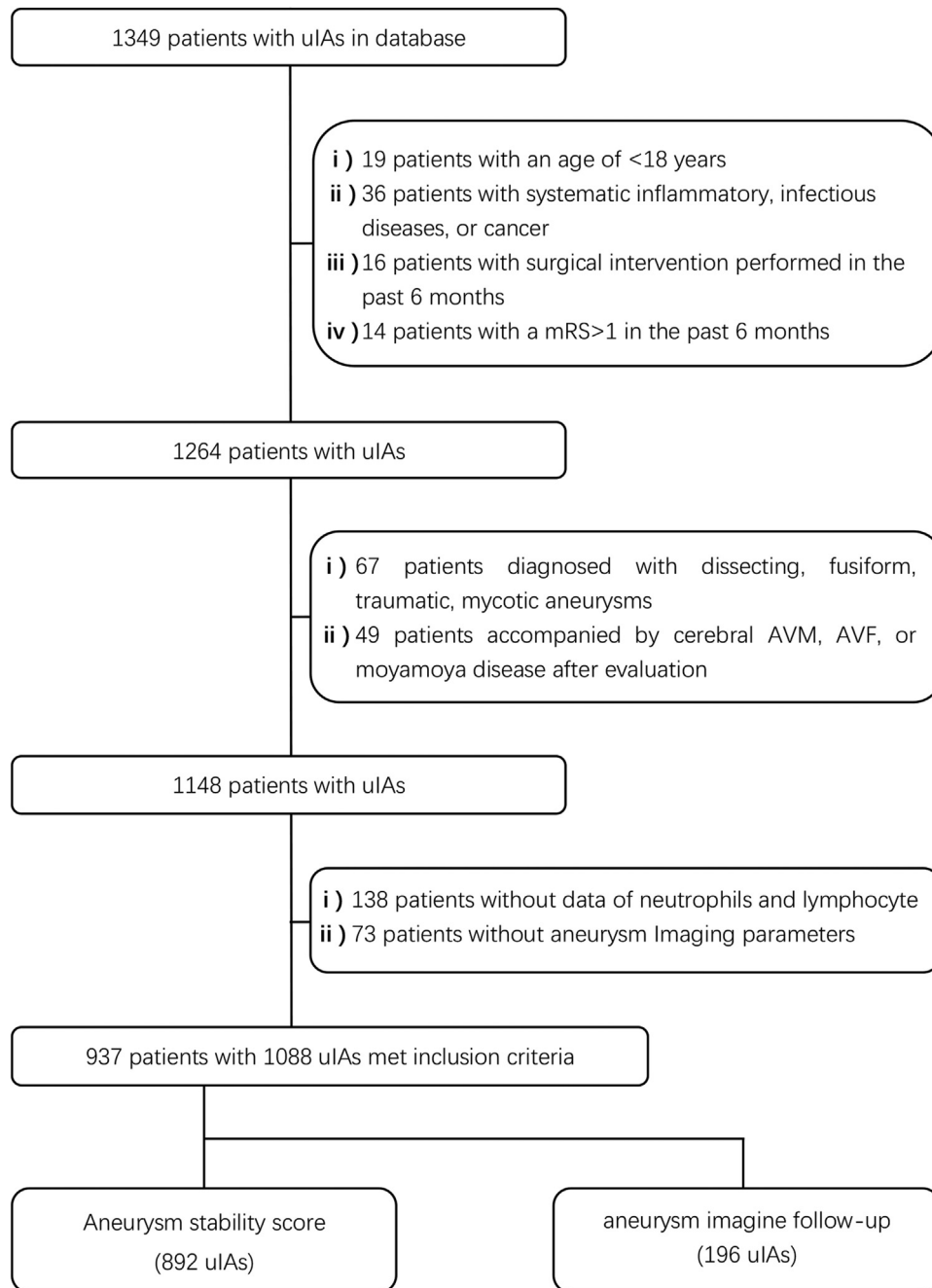


Fig. 1. Flow diagram of patient selection.

Abbreviations: uIAs, unruptured intracranial aneurysms; AVM = arteriovenous malformations; AVF = arteriovenous fistulas.

CI, 1.074–2.328], respectively), and JAPAN 3-year rupture risk score of >4 points (adjusted OR, 0.984 [95% CI, 0.670–1.444], 1.075 [95% CI, 0.733–1.576], and 1.512 [95% CI, 1.033–2.215], respectively).

Fig. 2A–C shows the non-linear relationship between the NLR and the risk of a high aneurysm stability score using logistic regression models with restricted cubic splines. We observed that a higher NLR was associated with an elevated risk of a higher aneurysm stability score. Fig. 3A–C shows the strength of the association between

each risk factor and high aneurysm stability scores (PHASES score of >5 points, ELAPSS score of >15 points, and JAPAN 3-year rupture risk score of >4 points).

Relationship between aneurysm growth and NLR

The median NLR in the aneurysm imagine follow-up group was 1.90 (interquartile range, 1.45–2.47). Supplemental Table 3 shows the baseline characteristics of

Table 1. Baseline characteristics of the aneurysms by neutrophil-to-lymphocyte ratio quartiles in the aneurysm stability score group.

Characteristics	Total aneurysms	Quartiles of neutrophil-to-lymphocyte ratio				P Value
		Q1, ≤1.416 (n=224)	Q2, 1.417-1.789 (n=222)	Q3, 1.790-2.390 (n=223)	Q4, ≥2.391 (n=223)	
Age, median (IQR), y	56.00 (48.00-62.00)	53.00 (47.00-60.00)	56.00 (49.00-62.00)	56.00 (49.00-62.00)	57.00 (49.00-63.00)	0.005
Female, n (%)	584 (65.5)	164 (73.2)	140 (63.1)	142 (63.7)	138 (61.9)	0.044
Medical history, n (%)						
SAH	14 (1.6)	2 (0.9)	2 (0.9)	6 (2.7)	4 (1.8)	0.411
Ischemic Stroke	155 (17.4)	23 (10.3)	44 (19.8)	44 (19.7)	44 (19.7)	0.015
Hypertension	440 (49.3)	88 (39.3)	104 (46.8)	119 (53.4)	129 (57.8)	0.001
Hyperlipidemia	110 (12.3)	19 (8.5)	30 (13.5)	32 (14.3)	29 (13.0)	0.232
Diabetes mellitus	85 (9.5)	10 (4.5)	21 (9.5)	22 (9.9)	32 (14.3)	0.005
Heart disease	72 (8.1)	10 (4.5)	21 (9.5)	18 (8.1)	23 (10.3)	0.111
Previous or current smoking, n (%)	179 (20.1)	38 (17.0)	42 (18.9)	59 (26.5)	40 (17.9)	0.050
Previous or current drinking, n (%)	200 (22.4)	43 (19.2)	54 (24.3)	54 (24.2)	49 (22.0)	0.524
Aneurysm morphological features						
Aneurysm size, median (IQR), mm	6.49 (4.42-11.17)	6.40 (4.33-11.42)	6.38 (4.46-11.15)	5.98 (4.30-10.05)	7.50 (4.69-12.98)	0.065
Maximum height, median (IQR), mm	5.52 (3.70-9.58)	5.38 (3.84-9.79)	5.50 (3.68-9.42)	5.24 (3.50-8.78)	6.12 (3.86-10.30)	0.289
Neck width, median (IQR), mm	4.36 (3.25-6.76)	4.22 (3.19-6.39)	4.36 (3.37-6.45)	4.37 (3.14-6.83)	4.67 (3.40-7.30)	0.281
Maximum width, median (IQR), mm	5.62 (3.70-9.69)	5.29 (3.66-9.18)	5.70 (3.63-9.95)	5.00 (3.55-8.92)	6.41 (3.94-10.87)	0.073
AR, median (IQR)	1.22 (0.97-1.70)	1.26 (1.00-1.64)	1.15 (0.97-1.66)	1.14 (0.92-1.67)	1.27 (0.97-1.92)	0.499
SR, median (IQR)	1.75 (1.14-2.99)	1.59 (1.06-2.99)	1.76 (1.13-2.89)	1.70 (1.12-2.90)	2.02 (1.30-3.18)	0.074
Parent vessel diameter, mean (SD), mm	4.02±1.31	4.04±1.29	4.13±1.50	3.92±1.21	4.01±1.23	0.595
Irregularity, n (%)	190 (21.3)	52 (23.2)	37 (16.7)	47 (21.1)	54 (24.2)	0.215
Daughter Sac, n (%)	129 (14.5)	38 (17.0)	20 (9.0)	34 (15.2)	37 (16.6)	0.060
Location, n (%)						
ICA	466 (52.2)	130 (58.0)	117 (52.7)	106 (47.5)	113 (50.7)	0.029
ACoMA	70 (7.8)	22 (9.8)	10 (4.5)	20 (9.0)	18 (8.1)	
PComA	102 (11.4)	24 (10.7)	18 (8.1)	29 (13.0)	31 (13.9)	
ACA	30 (3.4)	11 (4.9)	9 (4.1)	6 (2.7)	4 (1.8)	
MCA	65 (7.3)	13 (5.8)	22 (9.9)	17 (7.6)	13 (5.8)	
PC	159 (17.8)	24 (10.7)	46 (20.7)	45 (20.2)	44 (19.7)	
Aneurysm Stability Score, median (IQR)						
PHASES	5.00 (1.00-8.00)	4.00 (1.00-7.00)	4.00 (1.00-7.00)	5.00 (1.00-7.00)	5.00 (3.00-9.00)	0.011
ELAPSS	14.00 (8.00-22.00)	13.00 (6.25-22.00)	14.00 (8.00-22.00)	13.00 (7.00-22.00)	16.00 (10.00-23.00)	0.018
JAPAN 3-year rupture risk score	4.00 (2.00-7.00)	4.00 (1.00-7.00)	4.00 (2.00-6.50)	4.00 (2.00-7.00)	5.00 (2.00-7.00)	0.023
Laboratory parameter						
WBC, median (IQR), 10 ⁹ /L	6.35 (5.26-7.47)	5.59 (4.53-6.43)	6.25 (5.14-7.33)	6.45 (5.44-7.50)	7.01 (6.12-8.51)	<0.001
NEUT, median (IQR), 10 ⁹ /L	3.71 (2.87-4.62)	2.69 (2.17-3.10)	3.47 (2.89-4.18)	3.98 (3.27-4.64)	4.94 (4.18-6.01)	<0.001
LY, median (IQR), 10 ⁹ /L	1.99 (1.62-2.40)	2.32 (1.97-2.79)	2.21 (1.83-2.57)	1.94 (1.66-2.25)	1.54 (1.33-1.83)	<0.001
NLR, median (IQR)	1.79 (1.42-2.39)	1.19 (1.01-1.31)	1.58 (1.51-1.69)	2.03 (1.91-2.16)	2.93 (2.61-3.74)	<0.001

Abbreviations: IQR = interquartile range; Q = quartile; SAH = subarachnoid hemorrhage; AR = aneurysm maximum height to neck diameter ratio; SR = aneurysm maximum diameter to the parent vessel diameter(s) ratio; ICA = internal carotid artery; ACoMA = anterior communicating artery; PComA = posterior communicating artery; ACA = anterior cerebral artery; MCA = middle cerebral artery; PC = posterior circulation; WBC = white blood cell; NEUT = neutrophil; LY = lymphocyte; NLR = neutrophil-to-lymphocyte ratio.

Table 2. The quartile of neutrophil-to-lymphocyte ratio in the aneurysm stability score group and the risk of unstable aneurysms.

Aneurysm stability	Quartile of NLR	No.	Events, n (%)	Model 1* Adjusted OR (95% CI)	P Value	Model 2† Adjusted OR (95% CI)	P Value
PHASES >5	Q1, ≤1.416	224	82 (23.0)	Ref.		Ref.	
	Q2, 1.417-1.789	222	89 (25.0)	1.286(0.886-1.866)	0.186	1.267 (0.865-1.855)	0.225
	Q3, 1.790-2.390	223	83 (23.3)	1.525 (1.050-2.215)	0.027	1.490 (1.018-2.183)	0.040
	Q4, ≥2.391	223	102 (28.7)	2.128 (1.458-3.106)	<0.001	2.007 (1.361-2.960)	<0.001
	P for trend				<0.001		<0.001
ELAPSS >15	Q1, ≤1.416	224	86 (22.2)	Ref.		Ref.	
	Q2, 1.417-1.789	222	99 (25.5)	1.292 (0.885-1.884)	0.184	1.284 (0.874-1.887)	0.202
	Q3, 1.790-2.390	223	90 (23.2)	1.086 (0.743-1.587)	0.671	1.045 (0.709-1.539)	0.825
	Q4, ≥2.391	223	113 (29.1)	1.648 (1.132-2.401)	0.009	1.581 (1.074-2.328)	0.020
	P for trend				<0.001		0.001
JAPAN 3-year rupture risk score >4	Q1, ≤1.416	224	92 (23.7)	Ref.		Ref.	
	Q2, 1.417-1.789	222	89 (22.9)	0.960 (0.658-1.401)	0.833	0.984 (0.670-1.444)	0.933
	Q3, 1.790-2.390	223	94 (24.2)	1.046 (0.718-1.523)	0.817	1.075 (0.733-1.576)	0.712
	Q4, ≥2.391	223	114 (29.3)	1.501 (1.032-2.181)	0.033	1.512 (1.033-2.215)	0.034
	P for trend				<0.001		<0.001

Abbreviations: CI = confidence interval; OR = odds ratio; Ref = reference; Q = quartile; NLR = neutrophil-to-lymphocyte ratio.

*Model 1: No variable adjustment.

†Model 2: PHASES >5: adjusted for sex, history of ischemic stroke, hyperlipidemia, diabetes mellitus, heart disease, smoking status, drinking status, and aneurysm regularity; ELAPSS >15: adjusted for sex, history of ischemic stroke, hypertension, hyperlipidemia, diabetes mellitus, heart disease, smoking status, and drinking status; JAPAN >4: adjusted for history of ischemic stroke, hyperlipidemia, diabetes mellitus, heart disease, smoking status, drinking status, and subarachnoid hemorrhage history.

growth and non-growth aneurysms in aneurysm imagine follow-up group.

The baseline characteristics of the aneurysms in aneurysm imagine follow-up group by NLR quartile are shown in Table 3. In aneurysm imagine follow-up group, 39 (19.9%) aneurysms with growth. Table 4 shows the association between NLR quartiles and aneurysm growth. Aneurysms in the higher NLR categories were associated with aneurysm growth (*p* for trend <0.001). Compared with the lowest quartile, the second, third, and fourth NLR quartiles were associated with an increased risk of aneurysm growth (adjusted OR, 3.032 [95% CI, 0.497–18.513], 13.951 [95% CI, 2.597–74.942], and 16.759 [95% CI, 3.022–92.928], respectively).

Fig. 2D shows the non-linear relationship between the NLR and the risk of aneurysm growth using logistic regression models with restricted cubic splines. We observed that a higher NLR was associated with an elevated risk of aneurysm growth. Fig. 4 shows the strength of association between risk factors and aneurysm growth.

Incremental predictive value of NLR

We evaluated whether the NLR would further increase the predictive value of aneurysm growth. (Supplemental Table 4). The NLR did not significantly improve the C statistic of the PHASES score (C statistic 0.771–0.810, *p*=0.0785). However, the discriminatory power and risk reclassification seemed to be better (NRI across 5%–10%

cut points=19.7%, *p*=0.0028; continuous NRI=85.0%, *p*<0.0001; IDI=7%, *p*=0.0008). The NLR did not significantly improve the C statistic of the ELAPSS score (C statistic 0.777–0.807, *p*=0.2042). However, the discriminatory power was better (IDI=0.08%, *p*=0.0006). Risk reclassification results were as follows: NRI across 5%–10% cut points=12.7%, *p*=0.0757 and continuous NRI=79.9%, *p*<0.0001. The NLR significantly improved the C statistic, discriminatory power, and risk reclassification of the JAPAN 3-year rupture risk score (C statistic 0.790–0.832, *p*=0.0393; NRI across 5%–10% cut points=17.8%, *p*=0.0148; category-free NRI=81.1%, *p*<0.0001; IDI=7.3%, *p*=0.0018). For the prediction model established in this study, the NLR did not significantly improve the C statistic of the conventional model (C statistic 0.823–0.851, *p*=0.2666). However, the discriminatory power and risk reclassification seemed to be better (NRI across 5%–10% cut point=21.6%, *p*=0.0029; continuous NRI=87.5%, *p*<0.0001; IDI=9.2%, *p*=0.0002).

Discussion

Systemic inflammation may contribute the local inflammation reaction of IAs. Here, we show for the first time that aneurysms in the highest NLR quartile, compared with those in the lowest quartile, had a 2.007-fold higher risk of aneurysm instability as assessed by the PHASES score (>5), a 1.581-fold higher risk as assessed by the ELAPSS score (>15), and a 1.512-fold higher risk as

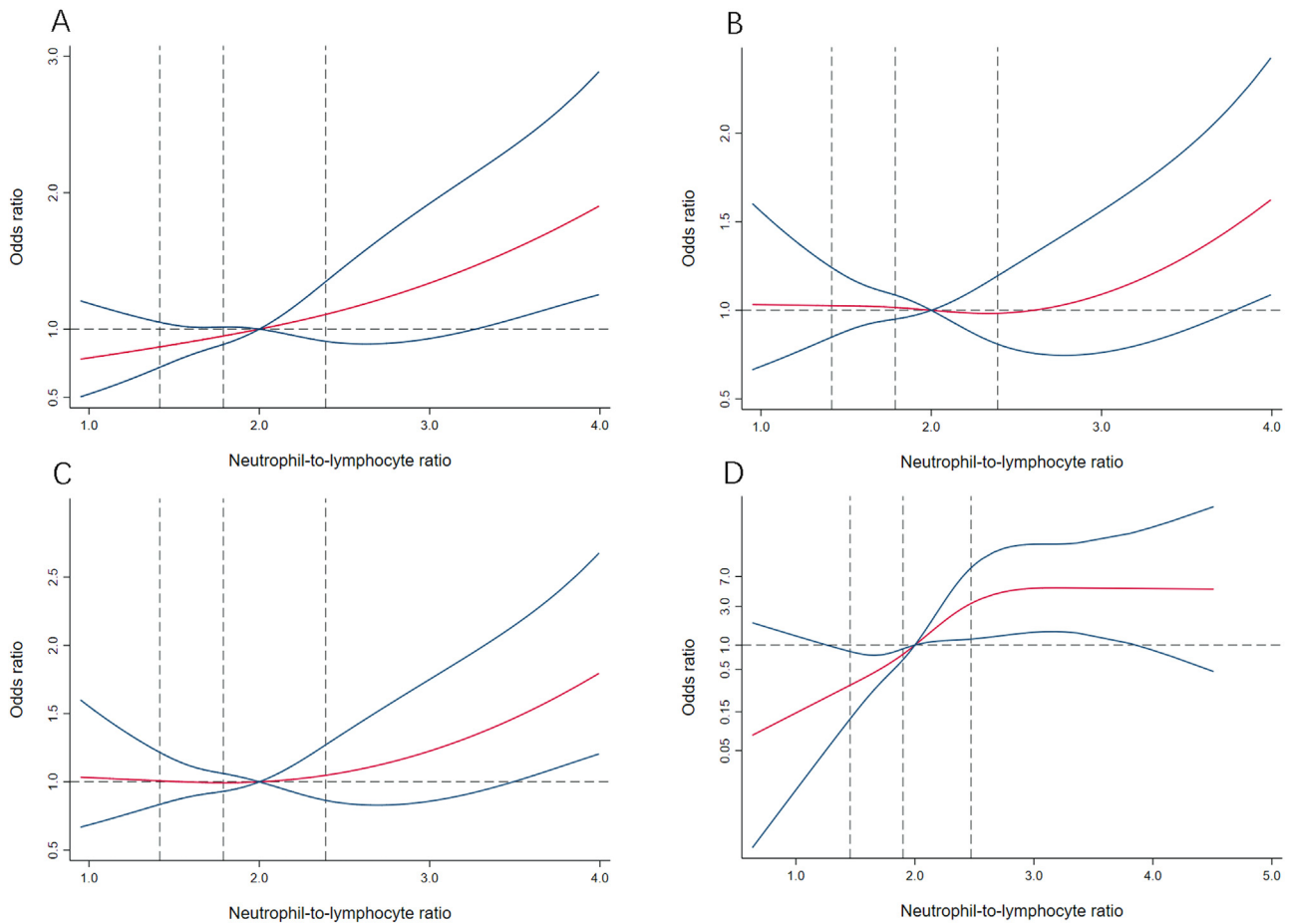


Fig. 2. Adjusted odds ratios of aneurysm stability score with PHASES score >5 points (A), ELAPSS score >15 points (B), and JAPAN 3-year rupture risk score >4 points (C) and aneurysm growth (D) according to neutrophil-to-lymphocyte ratio.

Red line indicates adjusted odds ratio, and blue lines indicate the 95% confidence interval bands. Reference is neutrophil-to-lymphocyte ratio = 2.00 for aneurysm stability score with PHASES score >5 points, ELAPSS score >15 points, and JAPAN 3-year rupture risk score >4 points and aneurysm growth. The other vertical dashed lines indicate the first, second, and third quartiles of neutrophil-to-lymphocyte ratio. The data were fitted using a logistic regression model of restricted cubic splines with 4 knots (25th, 50th, 75th, and 95th percentiles) for neutrophil-to-lymphocyte ratio, adjusting for potential covariates as model 2. The lowest 5% and highest 5% of participants are not shown in the figures because of small sample sizes.

assessed by the JAPAN 3-year rupture risk score (>4). The highest NLR quartile was significantly more closely associated with aneurysm growth (16.759-fold) than the lowest NLR quartile.

Aneurysm growth is related to pathological vascular remodeling caused by the recruitment and infiltration of inflammatory cells.²¹ The recruitment and infiltration of inflammatory cells further increases cytokine levels, matrix degradation, collagenase and protease concentrations, and apoptosis, leading to a decrease in the numbers of smooth muscle cells and weakening of the aneurysm structure; these changes ultimately result in aneurysm growth.^{8,9} Among these phenomena, the recruitment and infiltration of inflammatory cells (such as monocytes, macrophages, neutrophils, and lymphocytes) to the vessel wall is the key stage of aneurysm growth.¹⁰ Neutrophils are the first-responding inflammatory cells and migrate to the site of injury in the vascular wall. Studies of human IA

specimens have shown that an elevated concentration of plasma neutrophil elastase in patients with IAs is associated with vascular wall degeneration.^{22,23} In addition, the presence of neutrophilic enzymes in aneurysm walls is associated with a significantly increased risk of rupture, suggesting that neutrophils may play a causal role in the progression from aneurysm growth to rupture.¹¹ A recent study also showed that neutrophilic extracellular traps promote the development of IA rupture in animal aneurysm models.¹² Furthermore, neutrophils can secrete cytokines, such as tumor necrosis factor- α , CXC chemokine, macrophage inflammatory protein-1 α , and interleukin-1 β , and further promote the recruitment, migration, activation, and differentiation of inflammatory cells.²⁴ Our finding that higher numbers of circulating neutrophils are associated with aneurysm growth supports the notion that neutrophils promote the inflammatory progression of aneurysms from an epidemiological perspective.

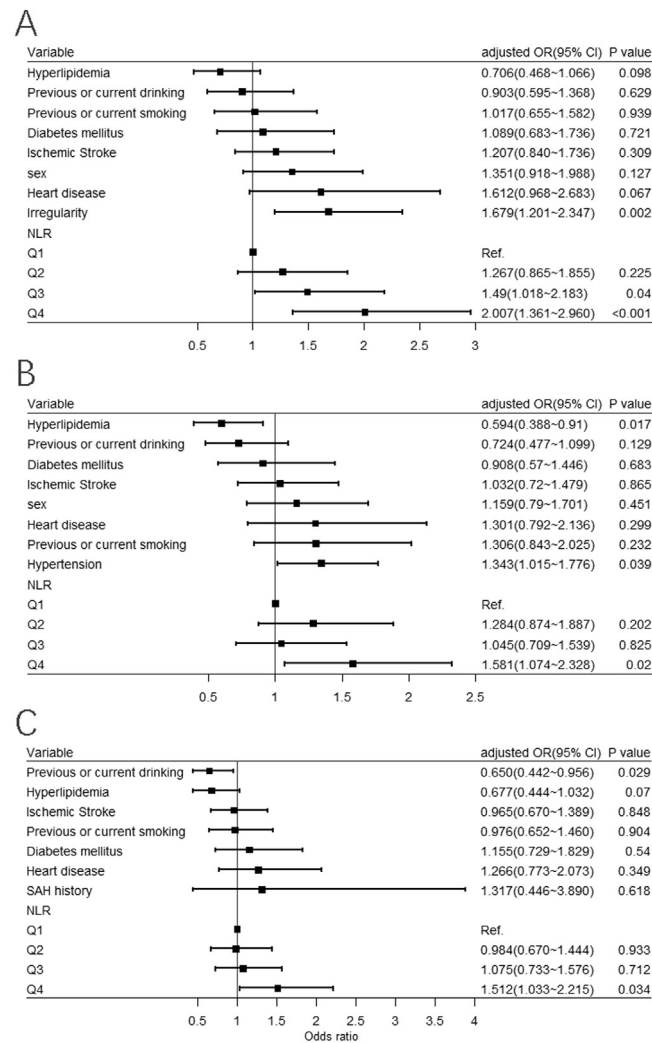


Fig. 3. Adjusted odds ratio of each risk factor for aneurysm stability evaluated by PHASES score >5 points (A), ELAPSS score >15 points (B), and JAPAN 3-year rupture risk score >4 points (C) with neutrophil-to-lymphocyte ratio. Abbreviations: CI = confidence interval; OR = odds ratio; Q = quartile; Ref = reference; NLR = neutrophil-to-lymphocyte ratio.

Lymphocytes are another major inflammatory cell found in aneurysms. T-cell infiltrates have also been found in human aneurysms, indicating that the immune system is widely involved in aneurysm growth.²⁵ In an animal model of elastase-induced aneurysms, T lymphocytes were observed to infiltrate the aneurysm wall.²⁶ T cells are the dominant cells in the aneurysm wall and are involved in the production of proinflammatory cytokines such as tumor necrosis factor- α , interferon- γ , and interleukin-6.¹⁰ Other studies have shown that lymphocyte-deficient Rag1 animals had a lower rate of aneurysm formation and rupture.¹⁵ In the present study, lymphocytes were not associated with aneurysm growth; however, this does not negate the role of lymphocytes in aneurysm progression. A previous study showed that infiltration of lymphocytes into the arterial wall occurs in the early stages of aneurysm formation.²⁷ In addition, that study used peripheral circulating lymphocyte counts, and no further lymphocyte typing was performed; this may

contribute to the difference between previous results and our results.

The NLR is the ratio of neutrophils to lymphocytes within the body and is a comprehensive index of nonspecific immunity and specific immunity.^{18,28-30} According to the pathological mechanism of aneurysm progression, the NLR seems to be related to the stability of aneurysm growth. Few epidemiological studies have focused on changes in the NLR in cerebral aneurysms. The present study is the first to show an association between the NLR and the stability of unruptured aneurysms. A recent study showed that the formation of abdominal aortic aneurysms is related to lymphocytes and neutrophils.¹⁶ However, the microenvironments differ between abdominal aortic aneurysms and IAs. Other studies also showed that the NLR is related to the severity of coronary artery disease.¹⁶

The NLR usually reflects the body's chronic inflammatory state.^{31,32} In this study, the NLR was related to aneurysm growth, supporting the hypothesis that chronic

Table 3. Baseline characteristics of the aneurysms by neutrophil-to-lymphocyte ratio quartiles in the aneurysm imagine follow-up group.

Characteristics	Total aneurysms	Quartiles of neutrophil-to-lymphocyte ratio				P Value
		Q1, ≤1.455 (n=49)	Q2, 1.456-1.900 (n=49)	Q3, 1.901-2.471 (n=50)	Q4, ≥2.472 (n=48)	
Age, median (IQR), y	60.00 (53.00-66.00)	59.00 (53.00-66.50)	60.00 (54.00-67.00)	60.00 (50.00-68.00)	59.50 (52.25-65.00)	0.836
Female, n (%)	130 (66.3)	34 (69.4)	33 (67.3)	31 (62.0)	32 (66.7)	0.885
Medical history, n (%)						
SAH	15 (7.7)	3 (6.1)	2 (4.1)	8 (16.0)	2 (4.2)	0.126
Ischemic Stroke	37 (18.9)	5 (10.2)	14 (28.6)	10 (20.0)	8 (16.7)	0.132
Hypertension	106 (54.1)	23 (46.9)	26 (53.1)	32 (64.0)	25 (52.1)	0.379
Hyperlipidemia	55 (28.1)	18 (36.7)	15 (30.6)	11 (22.0)	11 (22.9)	0.318
Diabetes mellitus	31 (15.8)	4 (8.2)	12 (24.5)	9 (18.0)	6 (12.5)	0.139
Heart disease	29 (14.8)	5 (10.2)	12 (24.5)	4 (7.8)	11 (21.2)	0.173
Previous or current smoking, n (%)	45 (23.0)	10 (20.4)	9 (18.4)	15 (30.0)	11 (22.9)	0.539
Previous or current drinking, n (%)	33 (16.8)	7 (14.3)	5 (10.2)	12 (24.0)	9 (18.8)	0.293
Aneurysm morphological features						
Aneurysm size, median (IQR), mm	4.50 (3.41-6.65)	4.40 (3.14-6.83)	4.64 (3.43-6.71)	4.91 (3.89-6.91)	4.07 (3.35-6.29)	0.292
Maximum height, median (IQR), mm	3.96 (2.67-5.58)	3.67 (2.42-5.71)	4.00 (2.75-5.95)	4.23 (3.27-5.85)	3.32 (2.56-4.58)	0.146
Neck width, median (IQR), mm	3.65 (2.86-4.56)	3.66 (2.81-4.45)	3.63 (2.66-4.81)	3.67 (2.95-4.55)	3.65 (3.00-4.48)	0.967
Maximum width, median (IQR), mm	3.57 (2.60-5.49)	3.57 (2.59-5.37)	3.59 (2.76-6.19)	4.03 (2.89-5.89)	3.24 (2.41-4.74)	0.185
AR, median (IQR)	1.06 (0.77-1.73)	0.94 (0.72-1.57)	1.19 (0.79-1.85)	1.11 (0.82-1.82)	0.99 (0.66-1.39)	0.309
SR, median (IQR)	1.33 (1.00-2.27)	1.24 (0.92-2.60)	1.47 (1.07-2.18)	1.46 (1.02-2.52)	1.20 (0.99-1.87)	0.514
Parent vessel diameter, mean (SD), mm	3.30±0.91	3.30±0.99	3.21±0.79	3.33±0.95	3.35±0.92	0.888
Irregularity, n (%)	34 (17.3)	7 (14.3)	11 (22.4)	6 (12.0)	10 (20.8)	0.455
Daughter Sac, n (%)	13 (6.6)	3 (6.1)	5 (10.2)	3 (6.0)	2 (4.2)	0.713
Location, n (%)						
ICA	122 (62.2)	30 (61.2)	33 (67.3)	28 (56.0)	31 (64.6)	0.174
ACoM	9 (4.6)	0 (0.0)	2 (4.1)	3 (6.0)	4 (8.3)	
PCoM	5 (2.6)	3 (6.1)	0 (0.0)	2 (4.0)	0 (0.0)	
ACA	6 (3.1)	2 (4.1)	4 (8.2)	0 (0.0)	0 (0.0)	
MCA	36 (18.4)	10 (20.4)	8 (16.3)	10 (20.0)	8 (16.7)	
PC	18 (9.2)	4 (8.2)	2 (4.1)	7 (14.0)	5 (10.4)	
Aneurysm Stability Score, median (IQR)						
PHASES	2.00 (1.00-5.00)	2.00 (0.50-5.00)	1.00 (0.00-5.00)	2.50 (1.00-5.00)	1.50 (0.00-5.00)	0.503
ELAPSS	7.00 (4.00-15.00)	7.00 (4.00-13.00)	8.00 (4.00-14.50)	9.50 (4.00-17.00)	5.50 (4.00-14.25)	0.734
JAPAN 3-year rupture risk score	2.00 (1.00-4.00)	2.00 (1.00-5.00)	2.00 (1.00-4.00)	3.00 (2.00-4.25)	2.00 (1.00-4.00)	0.677
Aneurysm growth, n (%)	39 (19.9)	3 (6.1)	6 (12.2)	16 (32.0)	14 (29.2)	0.002
Laboratory parameter						
WBC, median (IQR), 10 ⁹ /L	6.56 (4.91-7.80)	4.91 (4.34-6.61)	6.13 (5.11-7.46)	6.84 (5.21-7.92)	7.65 (6.74-8.82)	<0.001
NEUT, median (IQR), 10 ⁹ /L	3.83 (2.80-5.08)	2.47 (2.19-3.12)	3.52 (3.01-4.24)	4.17 (3.34-5.16)	5.31 (4.57-5.90)	<0.001
LY, median (IQR), 10 ⁹ /L	1.99 (1.64-2.31)	2.11 (1.81-2.80)	2.14 (1.68-2.50)	2.03 (1.62-2.30)	1.69 (1.52-2.00)	<0.001
NLR, median (IQR)	1.90 (1.45-2.47)	1.22 (0.98-1.33)	1.76 (1.58-1.85)	2.15 (1.98-2.28)	2.94 (2.66-3.31)	<0.001

Abbreviations: IQR = interquartile range; Q = quartile; SAH = subarachnoid hemorrhage; AR = aneurysm maximum height to neck diameter ratio; SR = aneurysm maximum diameter to the parent vessel diameter(s) ratio; ICA = internal carotid artery; ACoM = anterior communicating artery; PCoM = posterior communicating artery; ACA = anterior cerebral artery; MCA = middle cerebral artery; PC = posterior circulation; WBC = white blood cell; NEUT = neutrophil; LY = lymphocyte; NLR = neutrophil-to-lymphocyte ratio.

Table 4. The quartile of neutrophil-to-lymphocyte ratio in aneurysm imagine follow-up group and the risk of aneurysm growth.

Aneurysm stability Aneurysm growth	Quartile of NLR	No.	Events, n (%)	Model 1*	P Value	Model 2†	P Value
				Adjusted OR (95% CI) Ref.		Adjusted OR (95% CI) Ref.	
	Q1, ≤ 1.455	49	3 (7.7)				
	Q2, 1.456-1.900	49	6 (15.4)	2.140 (0.503-9.093)	0.303	3.032 (0.497-18.513)	0.229
	Q3, 1.901-2.471	50	16 (41.0)	7.216 (1.946-26.752)	0.003	13.951 (2.597-74.942)	0.002
	Q4, ≥ 2.472	48	14 (35.9)	6.314 (1.681-23.715)	0.006	16.759 (3.022-92.928)	0.001
	P for trend				<0.001		<0.001

Abbreviations: CI = confidence interval; OR = odds ratio; Ref = reference; Q = quartile; NLR = neutrophil-to-lymphocyte ratio.

*Model 1: No variable adjustment.

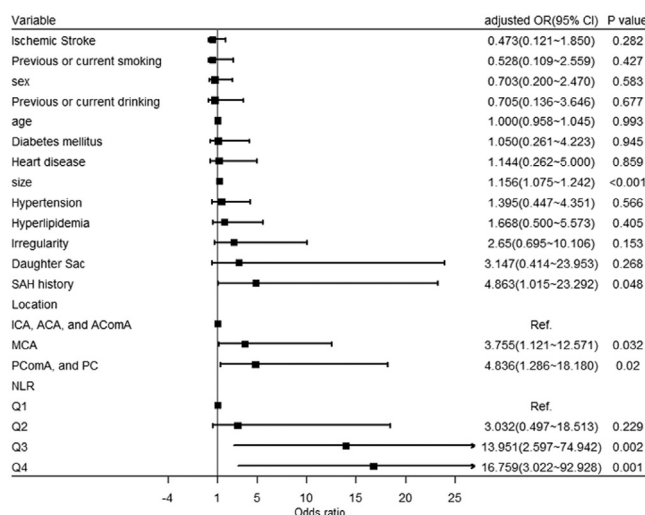
†Model 2: adjusted for age, sex, history of ischemic stroke, subarachnoid hemorrhage, hypertension, hyperlipidemia, diabetes mellitus, heart disease, smoking status, drinking status, size, location, daughter sac, and aneurysm regularity. Aneurysm location was divided into three groups according to the ELAPSS score: group_1: ICA, ACA, and AComA; group_2: MCA; group_3: PComA, and PC. Abbreviations: ICA = internal carotid artery; AComA = anterior communicating artery; PComA = posterior communicating artery; ACA = anterior cerebral artery; MCA = middle cerebral artery; PC = posterior circulation.

inflammation can promote the growth of aneurysms.^{33,34} Therefore, for patients with aneurysms, it is necessary to monitor the risk of aneurysm growth if the body is in a chronic inflammatory state as indicated by an increased NLR. In addition, this study showed a good correlation between the aneurysm stability score and the NLR. The aneurysm stability scores are cumulative scores based on the age, sex, hypertension and subarachnoid hemorrhage, aneurysm size and aneurysm shape, which also have been proven to be related to increases in the NLR.^{35–38} In addition, although diabetes mellitus, hyperlipidemia, cardiovascular disease, smoking, and other indicators are not included in the aneurysm stability score, these indicators have also been shown to be related to the growth and rupture of aneurysms.^{39–41} Moreover, in the present study, diabetes, hyperlipidemia, and smoking were

related to a high NLR in the univariate analysis. This indicates that the NLR is a comprehensive inflammatory indicator of risk factors for aneurysm growth and can be used to predict the risk of aneurysm growth.

Strengths and limitations

This is the first systematic study to report the association of the NLR with the stability and growth of unruptured aneurysms based on routine clinical data. However, we also acknowledge several limitations. First, the imaging follow-up of unruptured aneurysms was performed in selected patients with a lower risk of aneurysm growth. This may have resulted in underestimation of the overall risk of growth of unruptured aneurysms. Second, the observational study design did not allow us to establish a

**Fig. 4.** Adjusted odds ratio of each risk factor for aneurysm growth with neutrophil-to-lymphocyte ratio.

Aneurysm location was divided into three groups according to the ELAPSS score: group_1: ICA, ACA, and AComA; group_2: MCA; group_3: PComA, and PC.

Abbreviations: CI = confidence interval; OR = odds ratio; Q = quartile; Ref = reference; NLR = neutrophil-to-lymphocyte ratio; ICA = internal carotid artery; AComA = anterior communicating artery; PComA = posterior communicating artery; ACA = anterior cerebral artery; MCA = middle cerebral artery; PC = posterior circulation.

cause–effect relationship. Third, we did not evaluate the impact of the dynamic NLR on the results, and we lacked data on the distribution of lymphocyte subtypes (such as T or B cells), pro-inflammatory or anti-inflammatory subtypes, and the cell function/activation status. Fourth, because all patients in our study were Chinese, we do not know whether the results are applicable to other races. Fifth, because of the small data amount in the aneurysm growth assessment group, the NLR quartiles had high ORs and wide CIs; however, this did not affect the association between an elevated NLR and aneurysm growth. Finally, there might be some unknown confounding factors that we did not add to the analysis, and this may have also affected our results. Overall, however, we expect that our results will facilitate further studies exploring the growth of neutrophils and lymphocytes and aneurysms.

Conclusions

An elevated NLR was associated with an increased aneurysm stability score and aneurysm growth. This may help physicians to evaluate the stability and growth of unruptured IAs in clinical practice.

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Disclosures

None.

Declaration of Competing Interest

None.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jstrokecerebrovasdis.2023.107052](https://doi.org/10.1016/j.jstrokecerebrovasdis.2023.107052).

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